

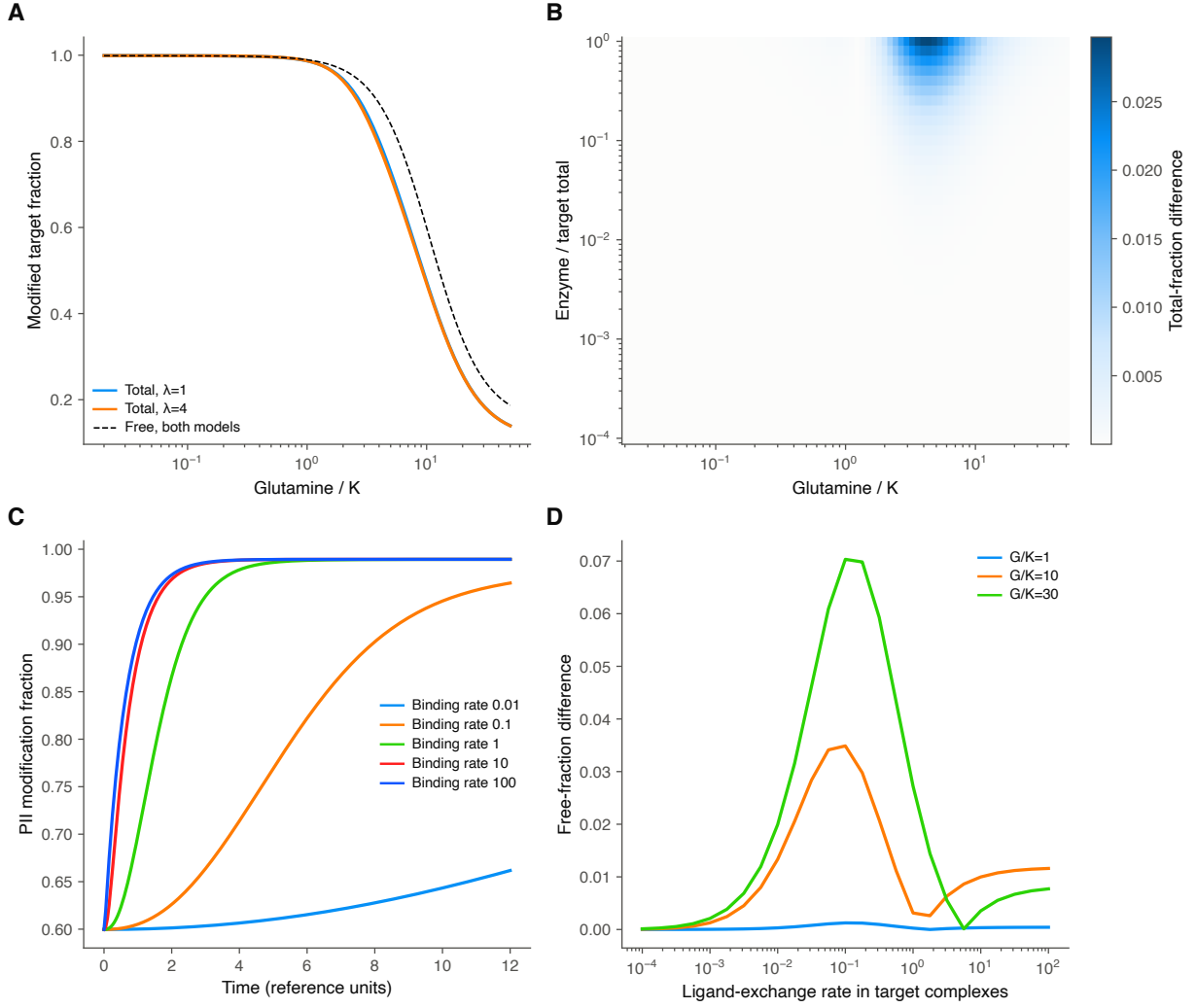


Figure S17: Reaction and observation assumptions define different boundaries. (A) Explicit enzyme–target complexes preserve the free modified fraction but change total fractions; enzyme/target total is 0.3. (B) Total-fraction separation between $\lambda = 1, 4$ across enzyme/target ratios for one disclosed asymmetric-affinity scenario. Conservation bounds this separation by $\min(1, E_{\text{tot}}/T_{\text{tot}})$ in the stated monovalent model. (C) Finite regulatory binding changes the transient without changing its steady endpoint. Binding rates and time are dimensionless reference units. (D) Ligand exchange in substrate-bound complexes with state-dependent dissociation/catalysis ratios breaks free-state equivalence. This extension preserves thermodynamic binding-cycle ratios but permits catalytic regulatory currents. Its parameters and positive steady-state checks are specified in S22. All curves are calculations, not new observations.